

ELEC 3115

Lecture 8: Week 4 , Thursday

217 responses submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)

field around conductors density and magnetic field strength
Magnetic flux field lines Magnetic dipoles magnetic saturation
field quantities **magnetic field** fields have similar
field and magnetic field intensity
field are formed electric currents electric fields ampere law
external fields postulates of magnetic Magnetic postulates
magnetic materials

217 responses submitted

What is still unclear or confusing to you right now? (Responses should reflect your own understanding.)

difference between b different material density and magnetic
vector potential field density magnetic dipole
poisson's equation **hysteresis** B and H magnetic potential
current density Hysteresis loops
field intensity Magnetic vector flux density magnetic field relation between B
field problems Relationship between B saturation or hysteresis

Revisiting: $\mathbf{B}$, $\mathbf{H}$, $\mathbf{M}$ relation (concept clarification)

- Orbiting electrons $\rightarrow$ microscopic current loop $\rightarrow$ magnetic dipole
- Magnetisation: Alignment of dipole moments with an applied magnetic field.
- Magnetisation vector $\mathbf{M}$: Net dipole moments per unit volume.

$$\vec{H} = \left(\frac{\vec{B}}{\mu_0} - \vec{M} \right) \quad \text{What does this mean?}$$

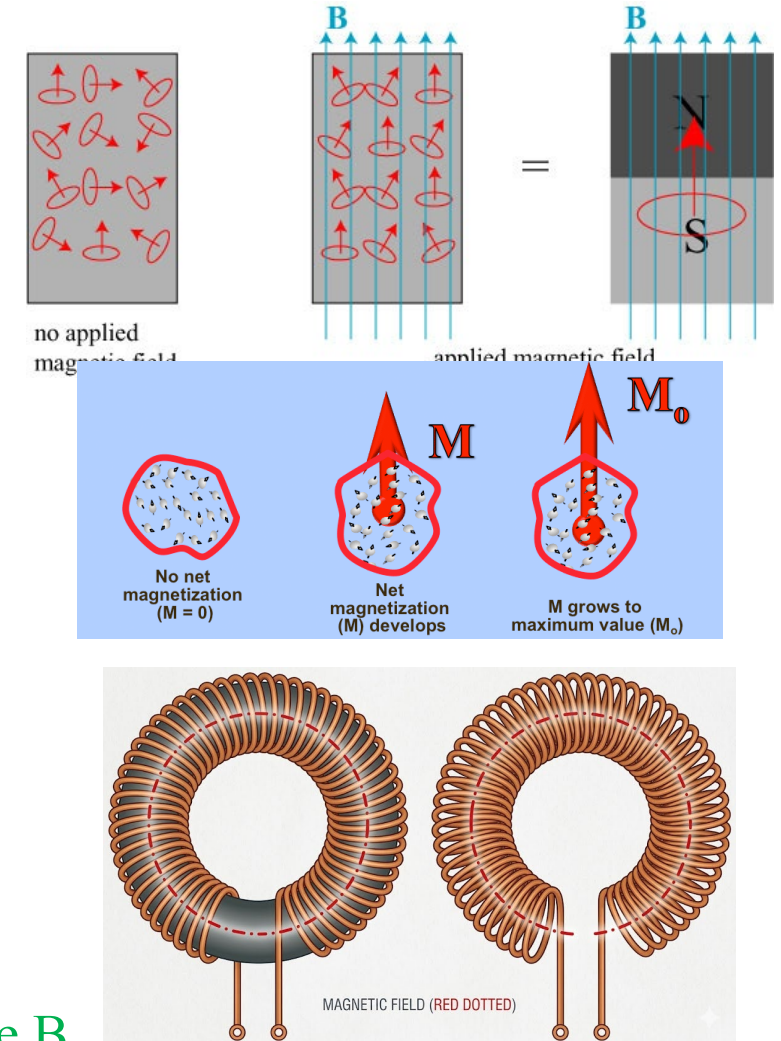
When $\mathbf{M}$ increases, $\mathbf{H}$ reduces: the more it is magnetised, the less effort (current) is required to establish the field.

Ampere's law in terms of $\mathbf{H}$ $\oint_C \vec{H} \cdot d\vec{l} = I_{enc}$

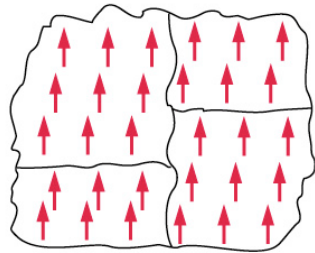
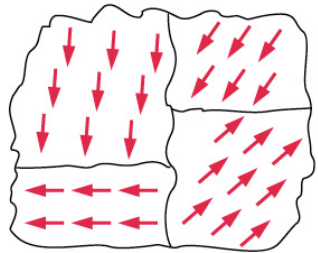
$\mathbf{H}$ is directly related to the current that induces a magnetic field.

When we have a magnetised material, the $\mathbf{H}$, and hence the current required to produce the same $\mathbf{B}$, reduces.

When the core material is magnetised, we need less current to produce the same $\mathbf{B}$.



Revisiting the Hysteresis Loop (Concept Clarification)



Ferromagnetism

What the Hysteresis or B–H Loop Really Means

When we apply a magnetic field H to a ferromagnetic material:

❶ Domains begin to align

→ Magnetic flux density B **increases**

❷ As H increases further

→ Alignment reaches the maximum

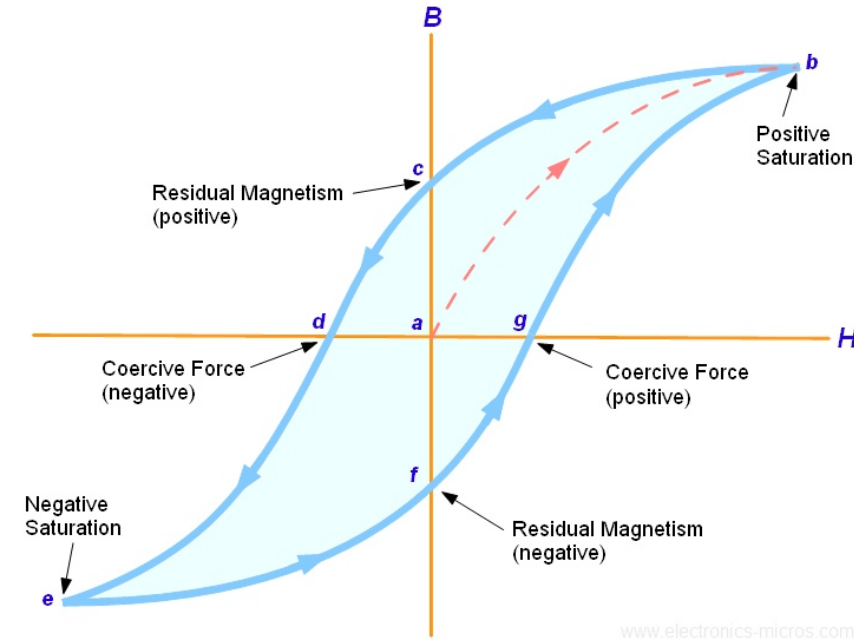
→ The curve flattens (approaching **saturation**)

❸ When we reduce H back to zero:

→ B **does NOT** return to zero (**remanence (B_r)**): material retains memory

→ Some domains remain aligned

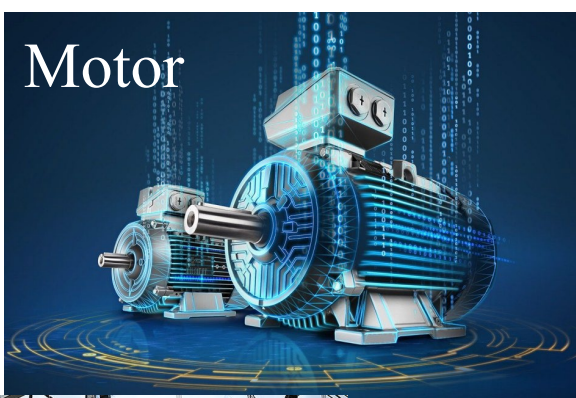
→ Need reverse H to make $B = 0$ (**coercivity (H_c)**)



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Generator



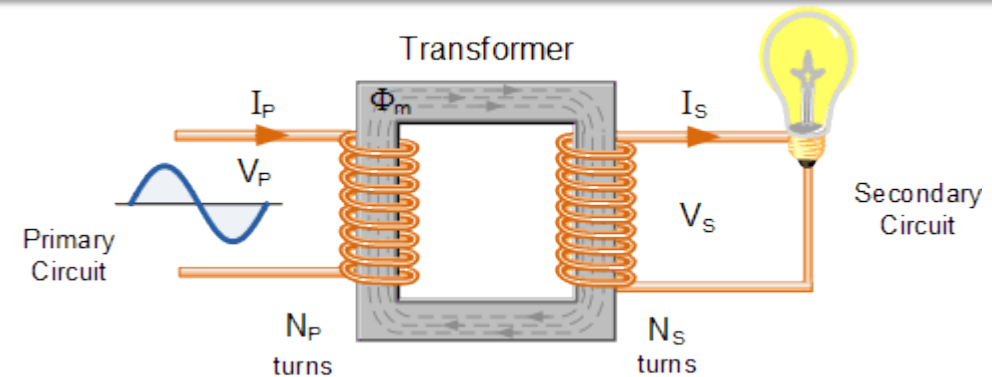
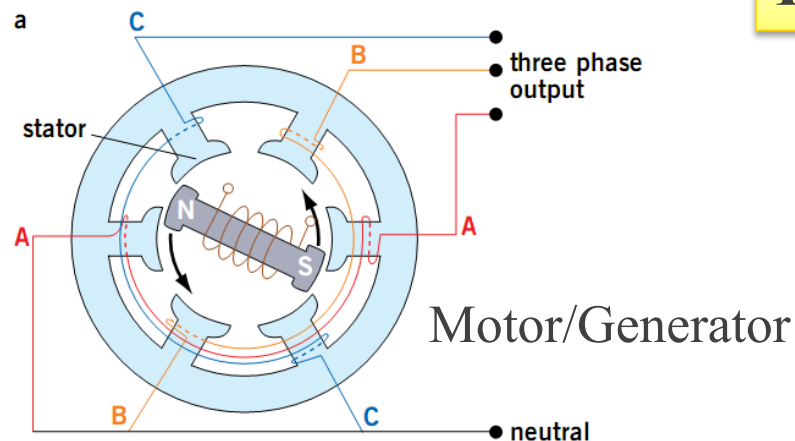
Motor



Transformer

How to analyze magnetic field in devices?

How to calculate inductance from magnetic field?



“Today’s Big Picture”

Magnetostatics (last lecture)



Magnetic Circuits (linear B-H relation)



Air Gaps & Reluctance



When μ is not constant (saturation) or known $\rightarrow$ Load-line method



Inductance formation (energy storage)



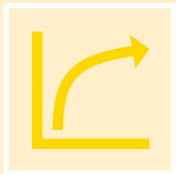
Design of solenoids and toroids

Learning goals

After today's lecture, you should be able to



Explain and calculate the key components of *magnetic circuit* – reluctance, flux and magnetomotive force (MMF).



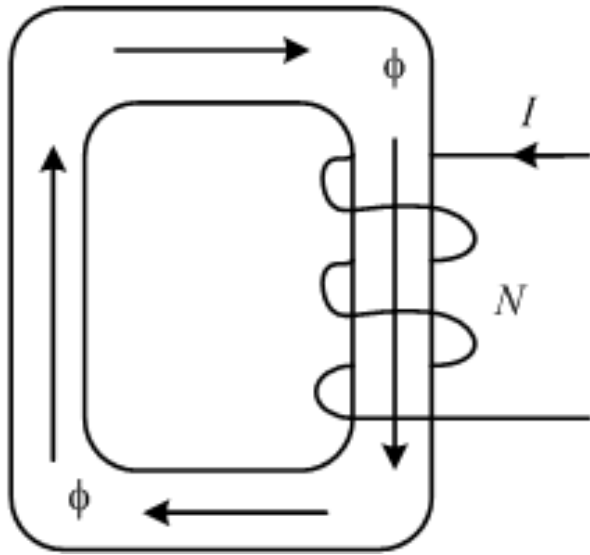
Identify *limitation* of magnetic circuit and use alternative *graphical method* to analyze magnetic devices when required.



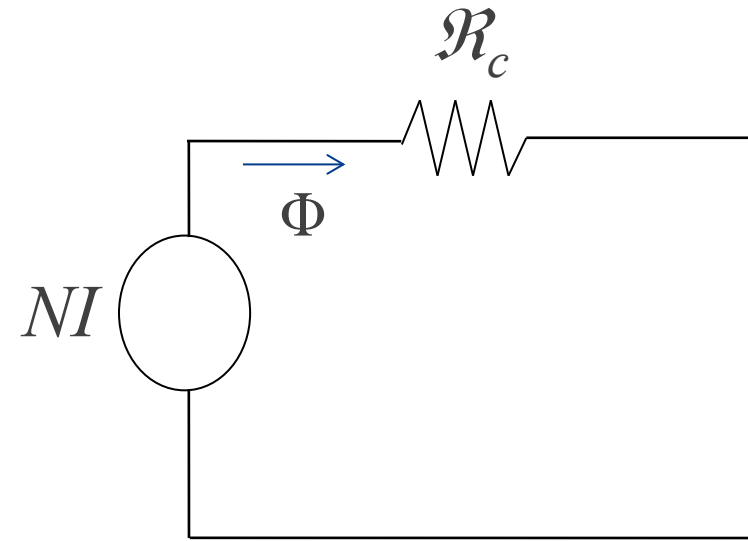
Explain the fundamentals behind inductance formation.
Calculate and design solenoid/toroid inductors.

Magnetic circuit representation

A magnetic device can be analysed using the concepts of circuit *if there is no magnetic saturation*.



Magnetic device



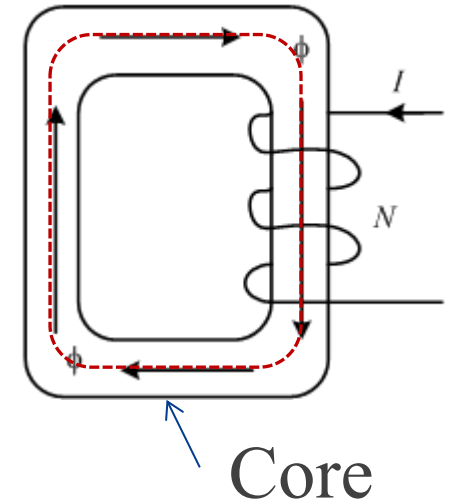
Magnetic circuit

Reluctance and MMF

Let's consider flux ϕ is being induced in an iron core by passing a current I through a coil of N turns wound on the core as shown in the figure.

According to Ampere's Law,
$$\oint_c \vec{H} \cdot d\vec{l} = I_{\text{enclosed}}$$

Can you identify the closed contour C in the circuit? What is I_{enclosed} here?



We start from Ampere's law,

$$\oint_c \vec{H} \bullet d\vec{l} = I_{enclosed} \rightarrow Hl_c = NI$$

$$\because B = \mu H, \quad \because \Phi = BS_c,$$

$$\text{sub H } \frac{B}{\mu} l_c = NI \longrightarrow \text{sub B, } \frac{\Phi}{\mu S_c} l_c = NI$$

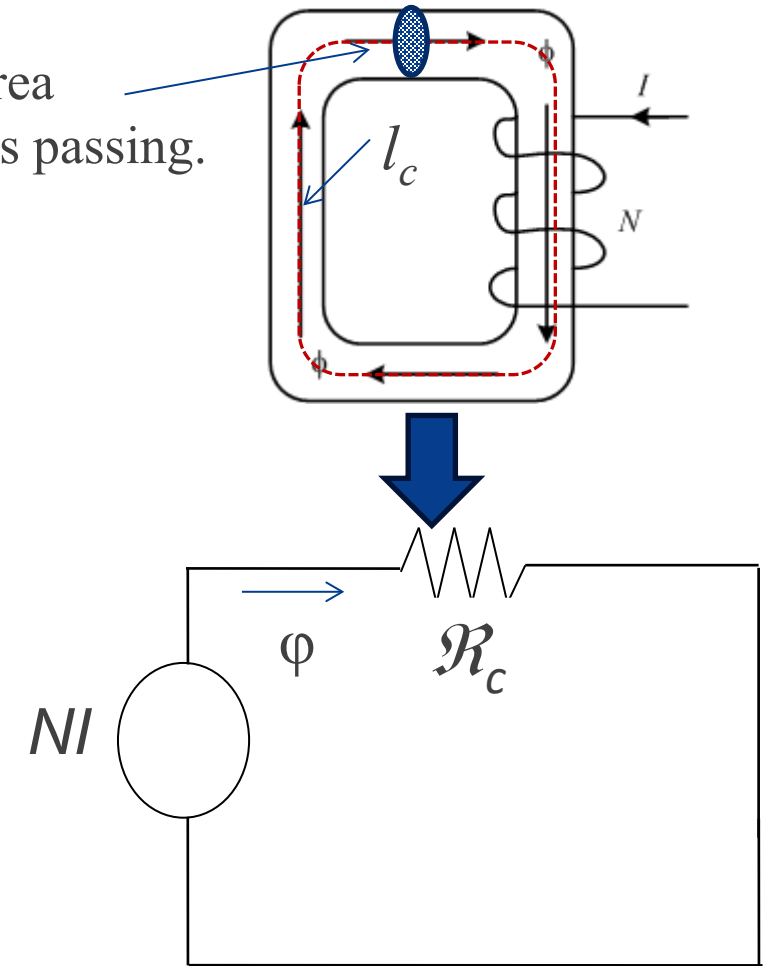
We can define, $\mathfrak{R}_c = \frac{l_c}{\mu S_c} \rightarrow \text{reluctance.}$

$$\therefore \mathfrak{R}_c \Phi = NI$$

Flux ϕ is induced by NI , which is referred as magnetomotive force (MMF). Unit is Ampere-turns (A-t).

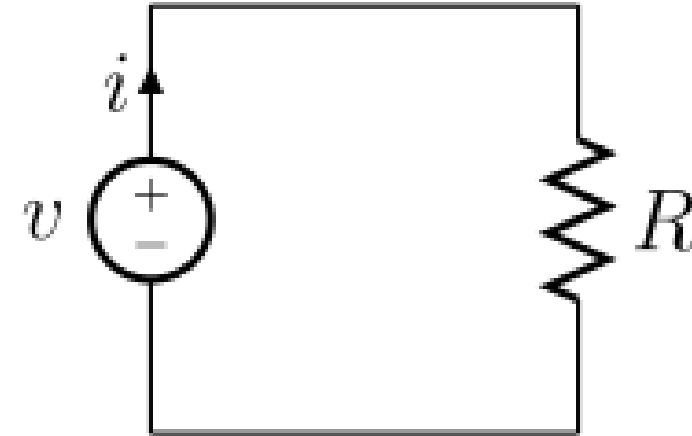
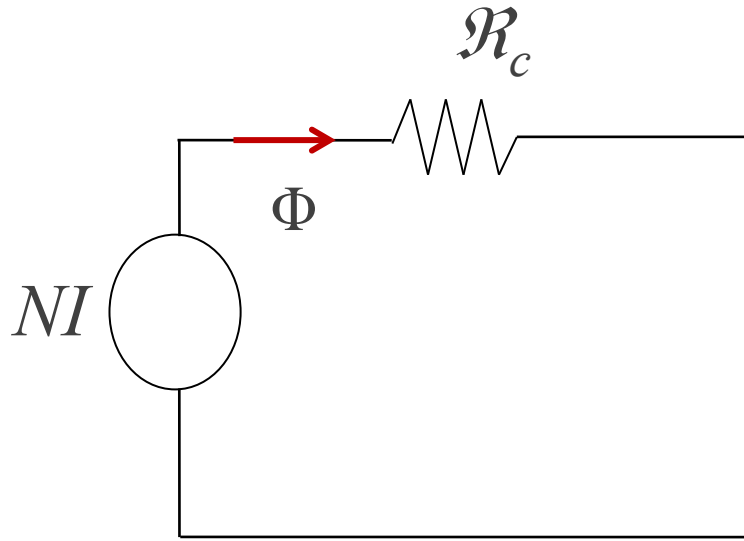
Reluctance is analogous to resistance of electric circuit and unit is A-t/Wb.

S_c : cross-sectional area through which flux is passing.



$$\Phi = \frac{NI}{\mathfrak{R}_c}$$

EXAM skill



Magnetic circuit	Electric Circuit
Magnetomotive force (MMF) NI	Electromotive force (EMF) v
Magnetic flux Φ	Electric Current i
Reluctance $\mathcal{R}$	Resistance R
Permeability μ	Conductivity σ

- Magnetic circuit method works only in linear region (constant μ).

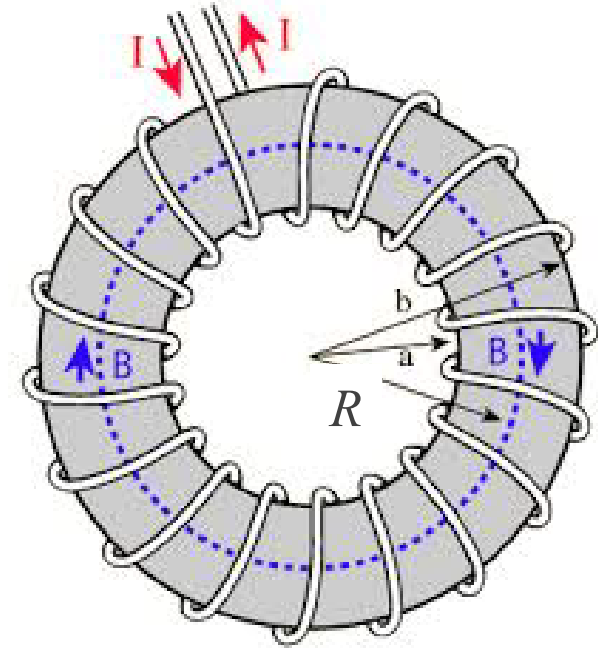
Example 2

1. A toroidal coil's cross-sectional radius is $R_0=1\text{m}$ and the mean radius of the torus is $R=1\text{km}$.

(i) Find the reluctance of the torus core if it is made of magnetic steel of $\mu_r=5000$.

(ii) If $N=100$ and $I=100\text{A}$, find the flux in the core made of magnetic steel.

Given: $\mu_0=4\pi \times 10^{-7} \text{ H/m}$



$$(i) \mathcal{R}_c = \frac{l_c}{\mu S_c}, \quad \mu = \mu_0 \mu_r$$

$$\text{Ans: } 3.2 \times 10^5 \text{ At/Wb}$$

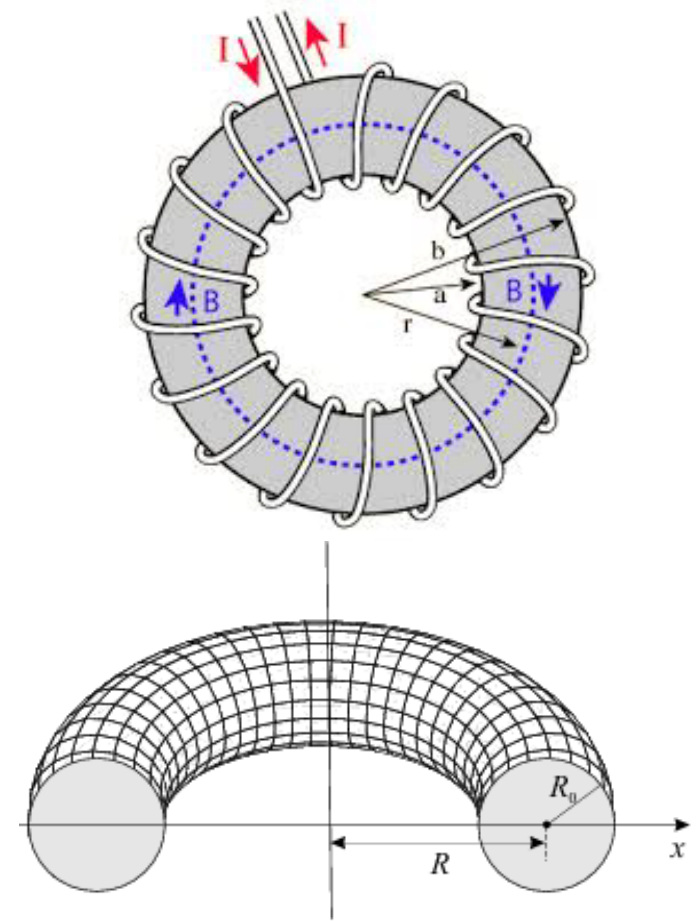
$$(ii) \Phi = \frac{NI}{\mathcal{R}_c}$$

$$\text{Ans: } 0.0314 \text{ Wb}$$

$$\text{Recall: } B = \frac{\mu_0 NI}{2\pi r} \text{ where } a < r < b$$

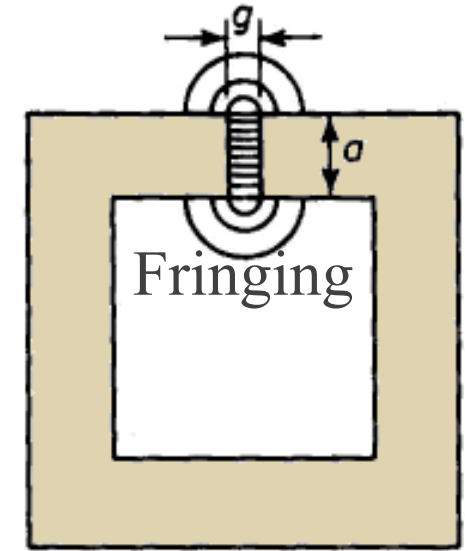
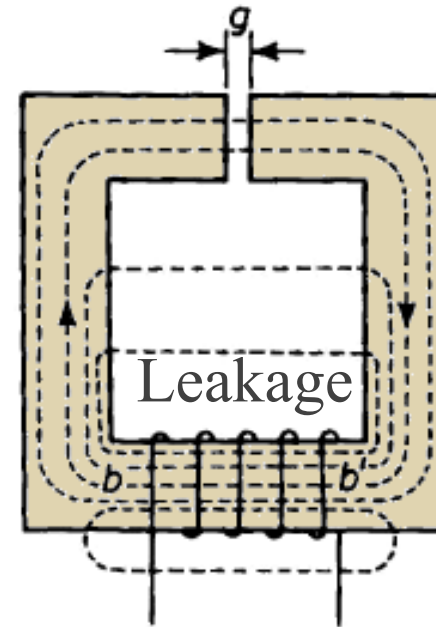
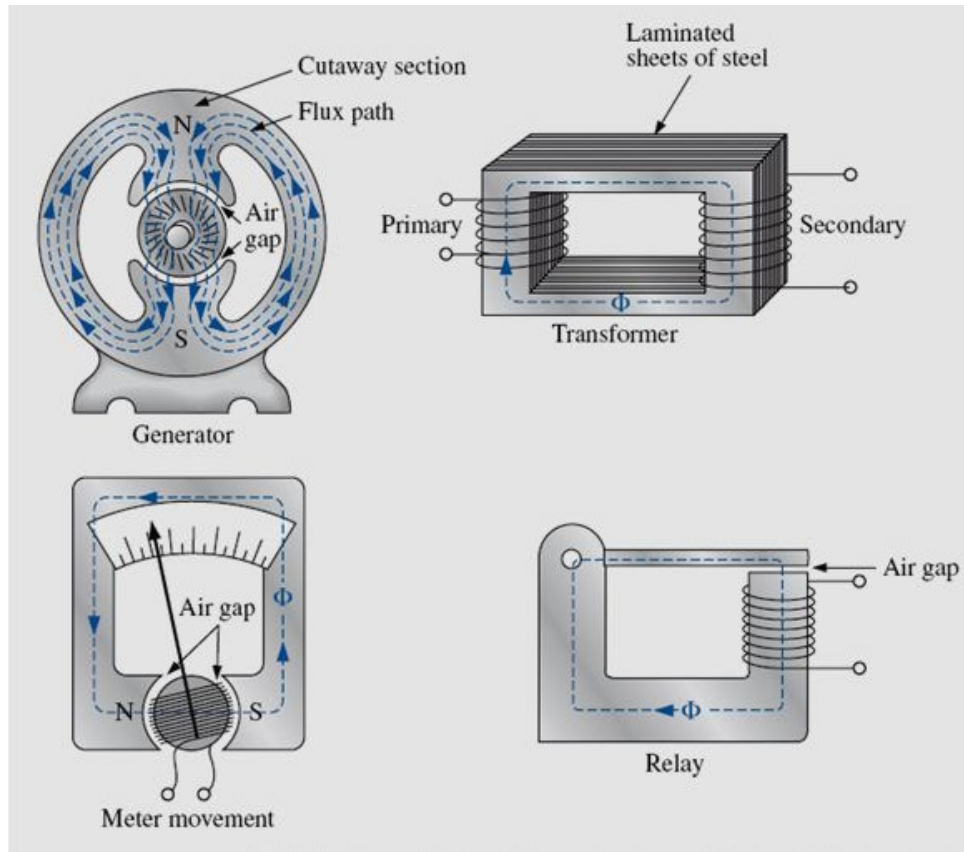
Flux density varies with r . The magnetic circuit assumes B is uniform in the cross-section and uses a mean radius. This assumption is valid when a and b are larger than the radial thickness ($b-a$).

Two key **assumptions** (i) B is uniform in the core (ii) no saturation (i.e., permeability is constant and known).



Magnetic devices with air-gaps

Many magnetic devices have air-gaps along the flux path.



Assumptions

Magnetic devices with air-gap can be analysed using magnetic circuit if (i) there is no significant *leakage* and *fringing* (ii) B is uniform and (iii) no saturation.

Magnetic circuit representation of air-gap

According to Ampere's Law,

$$\oint_c \vec{H} \cdot d\vec{l} = NI \rightarrow H_c l_c + H_g l_g = NI$$

$$\frac{B_c}{\mu_c} l_c + \frac{B_g}{\mu_0} l_g = NI \quad \text{Note : } \mu_c = \mu_0 \mu_r$$

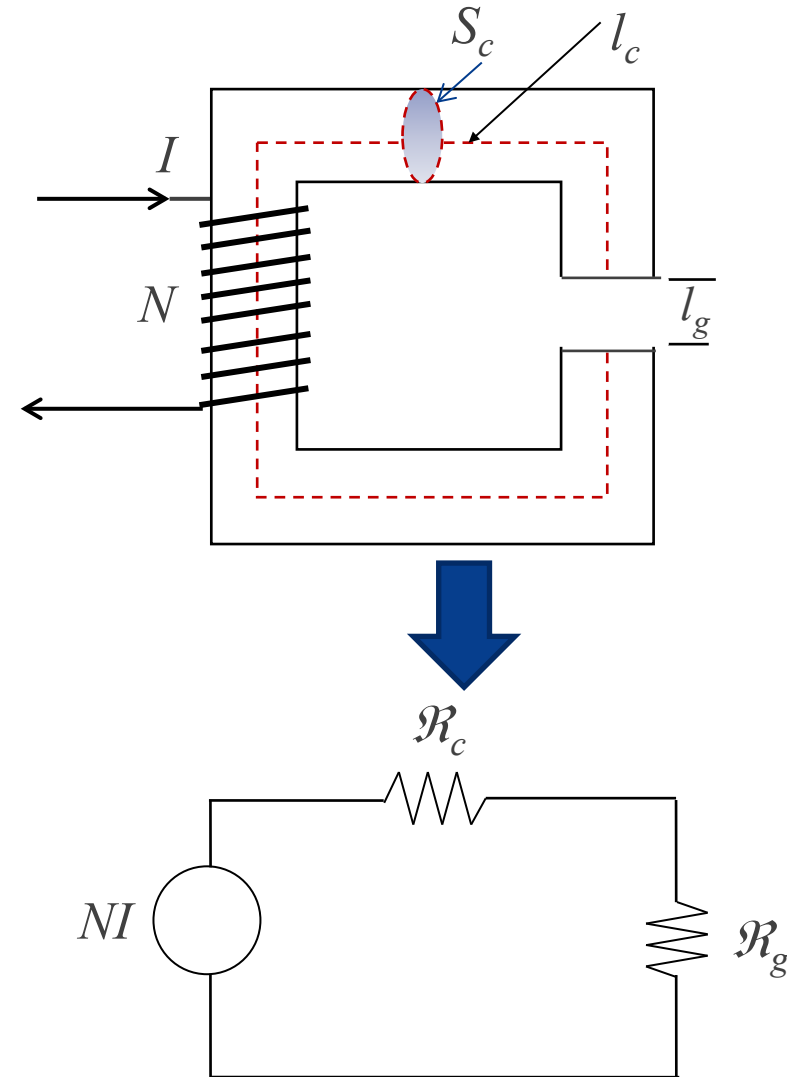
$$\frac{\Phi_c}{S_c \mu_c} l_c + \frac{\Phi_g}{S_g \mu_0} l_g = NI$$

$$\therefore \mathcal{R}_c \Phi_c + \mathcal{R}_g \Phi_g = NI$$

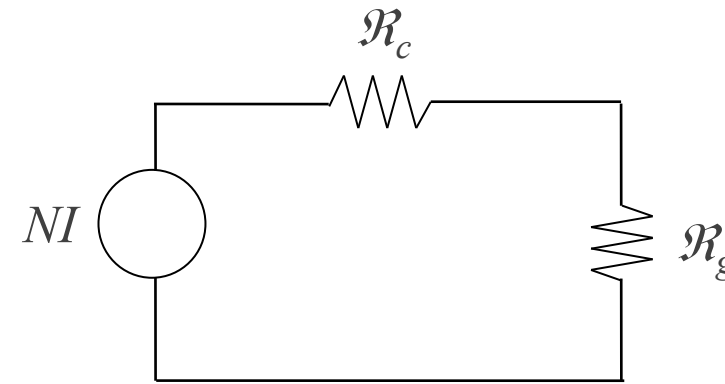
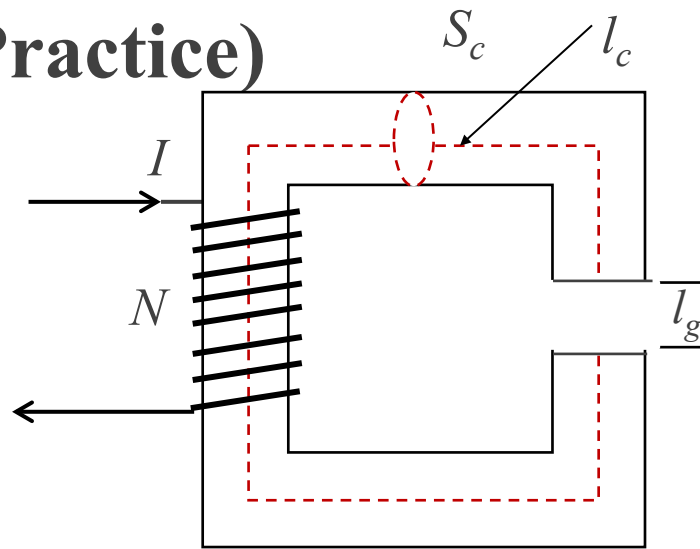
If no leakage flux, $\Phi_c = \Phi_g = \Phi$

If no fringing, $S_c = S_g$

$$\therefore \Phi = \frac{\text{Applied MMF}}{\text{Total Reluctance}} = \frac{NI}{(\mathcal{R}_c + \mathcal{R}_g)}$$



Example 3 (Practice)



The magnetic circuit shown in Fig above has dimensions $S_c = S_g = 9 \text{ cm}^2$, $l_g = 0.050 \text{ cm}$, $l_c = 30 \text{ cm}$ (length of the red dotted line), and $N = 500$ turns. The coils is carrying current $I = 0.8 \text{ A}$. Assume the value $\mu_r = 70,000$ for the core material. Calculate the two reluctances and then find the flux density in the core assuming there is no leakage and fringing. $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ **Ans: 1T.**

$$\mathcal{R}_c = \frac{l_c}{S_c \mu_c} \quad \mathcal{R}_g = \frac{l_g}{S_g \mu_0}$$

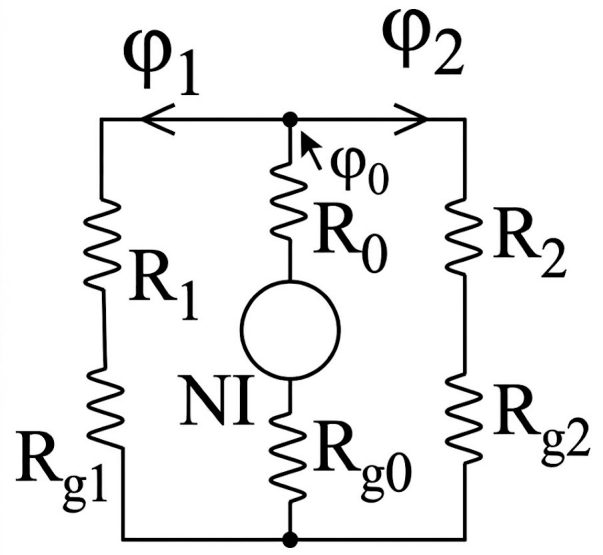
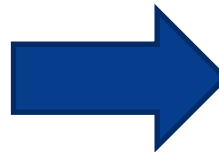
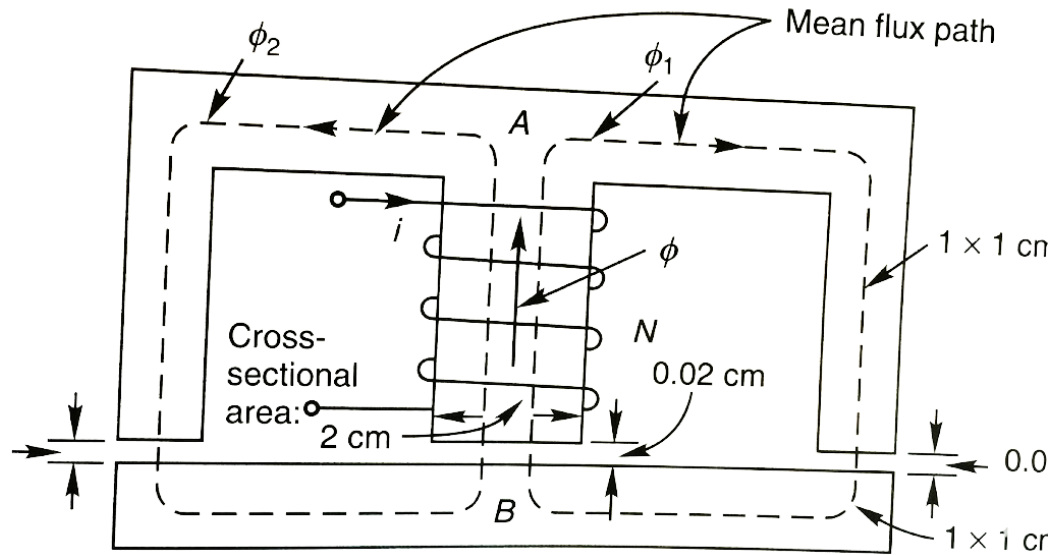
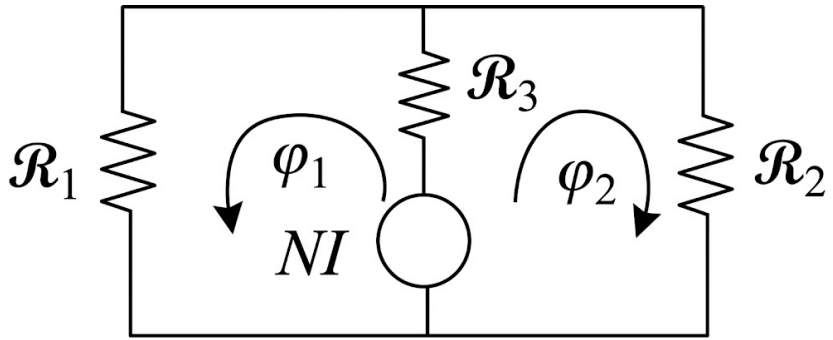
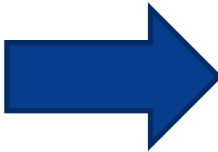
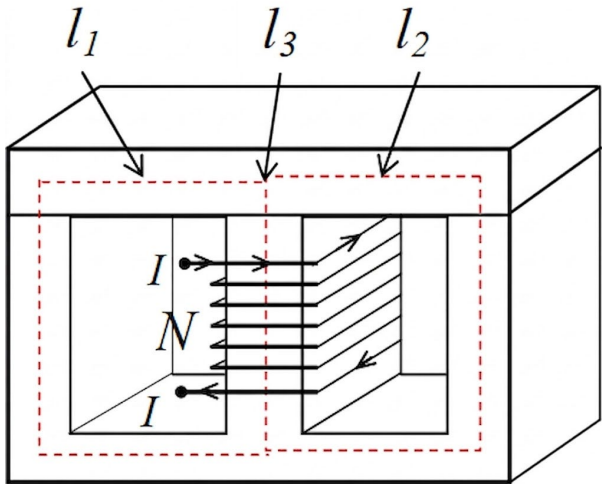
$$\Phi = \frac{NI}{(\mathcal{R}_c + \mathcal{R}_g)}$$

Comparing the reluctance of the airgap to the core, what can you conclude? The core with large relative permeability is sometimes referred to as the *infinitely permeable*. What is the consequence of this?

● In most magnetic devices, the air gap controls the flux.

More examples

$$\mathfrak{R} = \frac{l}{\mu S}$$



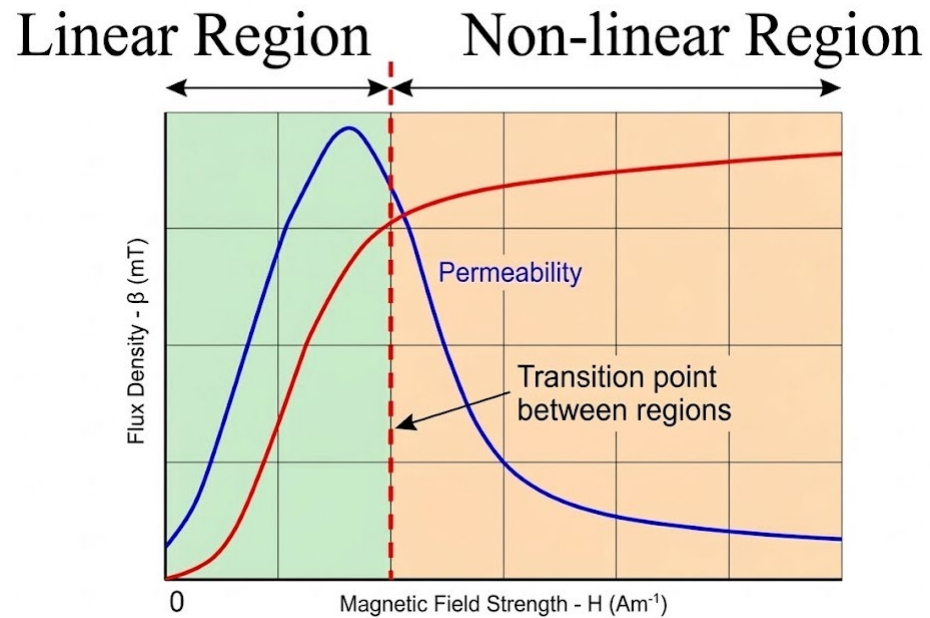
When Does Magnetic Circuits become inapplicable?

$$\mathfrak{R} = \frac{l}{\mu S}$$

When μ is not known or not constant, reluctance cannot be calculated.

When magnetic material saturates, i.e. operation in the non-linear region, μ varies and the flux cannot be calculated using,

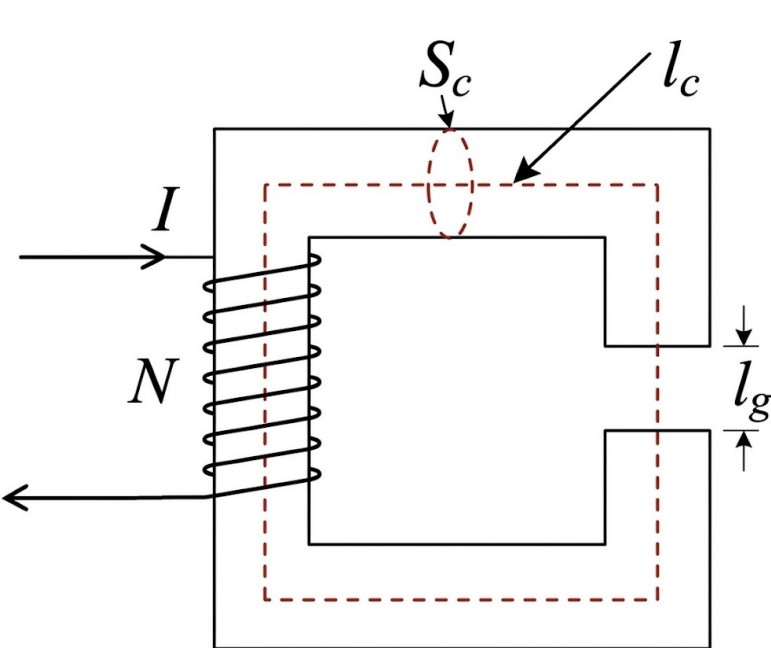
$$\Phi = \frac{NI}{\mathfrak{R}}$$



Graphical solution for B and H

Graphical solution or load-line method is an alternative to the magnetic circuit method, when the value of μ is not known or not a constant (i.e. core is saturating) – a graphical way to solve a non-linear B-H core material problem.

Let's consider the same magnetic circuit as before. But consider core permeability μ_c is not known.



$$\oint_c \vec{H} \cdot d\vec{l} = I_{enc} \rightarrow H_c l_c + H_g l_g = NI$$

$$\text{Sub } H_g, \quad H_c l_c + \frac{B_g}{\mu_0} l_g = NI$$

Multiply both sides by μ_0/l_g ,

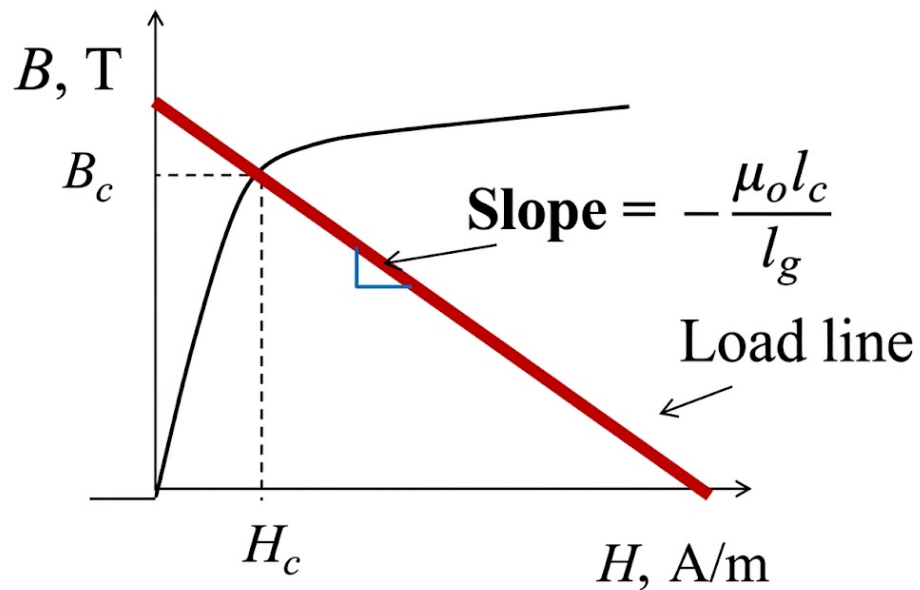
$$B_g + \mu_0 \frac{l_c}{l_g} H_c = \frac{\mu_0}{l_g} NI$$

$$\text{Rearrange, } B_g = \frac{\mu_0}{l_g} NI - \mu_0 \frac{l_c}{l_g} H_c$$

$$B_g = \frac{\mu_0}{l_g} NI - \mu_0 \frac{l_c}{l_g} H_c$$

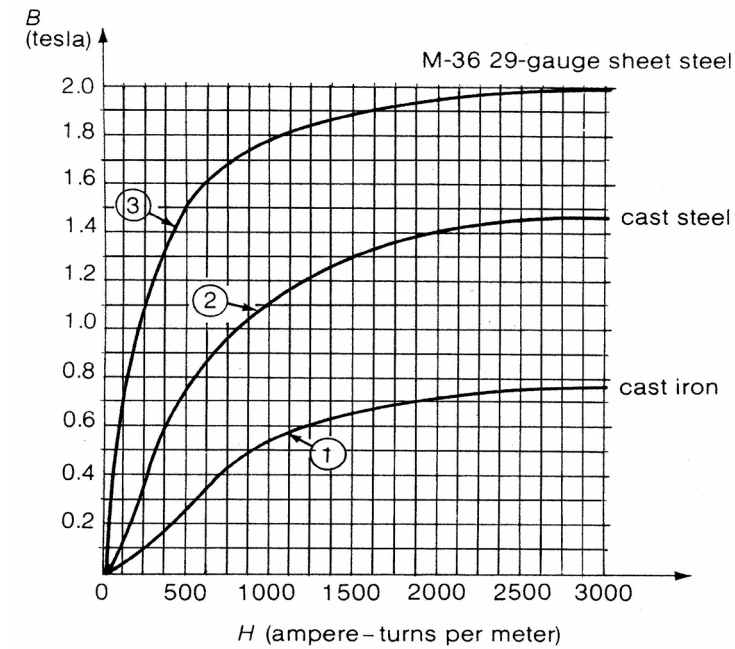
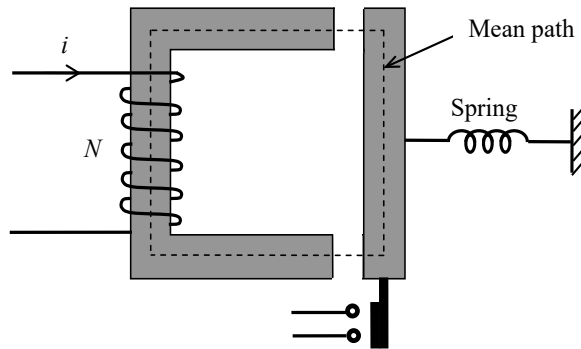
In a general form: $B = \frac{\mu_0}{l_g} NI - \mu_0 \frac{l_c}{l_g} H$

This equation has the form of $y = c - mx$ i.e., equation of a straight line (called load line). By superimposing the load line on the magnetization or BH curve of the core material, the operating H_c and B_c can be found.



So, for load-line method we must know the B-H (or magnetization) curve of the core material.

Example 4 (Application)



Consider the magnetic circuit of a relay as shown in Figure (a). The coil has 1200 turns, and the mean core length is $l_c = 360$ mm. The air gap lengths are 1.5 mm each. The air gap and the core have a uniform cross section of 2000 mm^2 . A flux density (B) of 0.8 Tesla in the air gap is required for the relay to close (i.e., actuate). The magnetic material of the core of the relay is cast steel (see curve 2)

(i) Find the minimum current required in the coil for the relay to actuate.

(ii) With the relay closed, the air gap is reduced to 0.5 mm. If the coil current stays at the same value found in (i), find the B field in the air gap.

In this example, do we need to use the graphical method? Why or why not?

$$B = \frac{\mu_0}{l_g} NI - \mu_0 \frac{l_c}{l_g} H$$

Pause & Reflect

(Before we take a short break...)

 **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

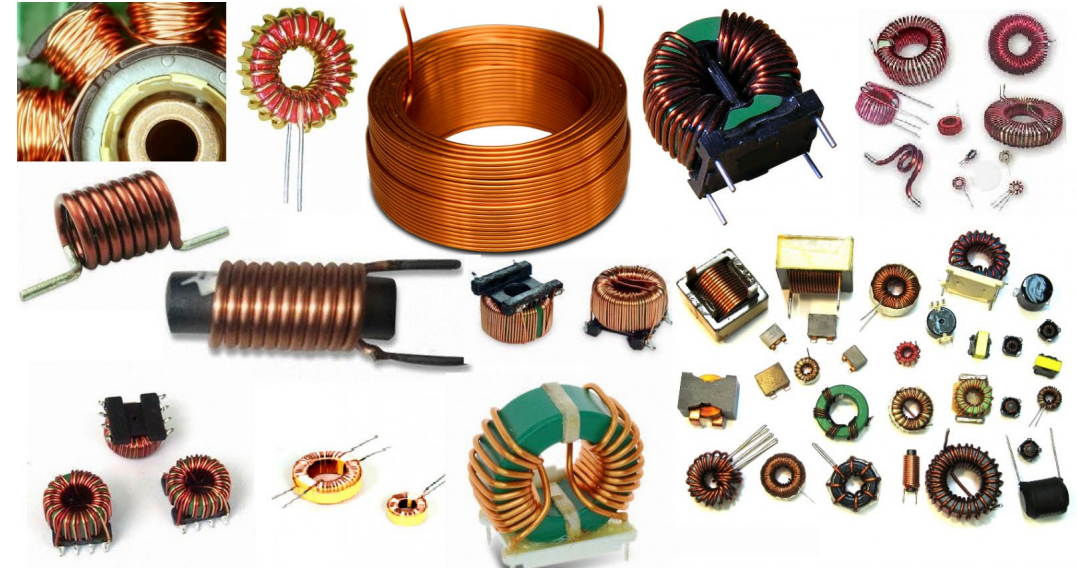
One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

💡 *This is for you — no submission.*



Inductance formation

Let's us recall what we know about inductance



Inductance L : What is inductance physically?

Current I in the loop produces magnetic flux density $\mathbf{B}$. $\oint_c \vec{B} \bullet d\vec{l} = \mu I_c$

$\mathbf{B}$ is directly proportional to current I .

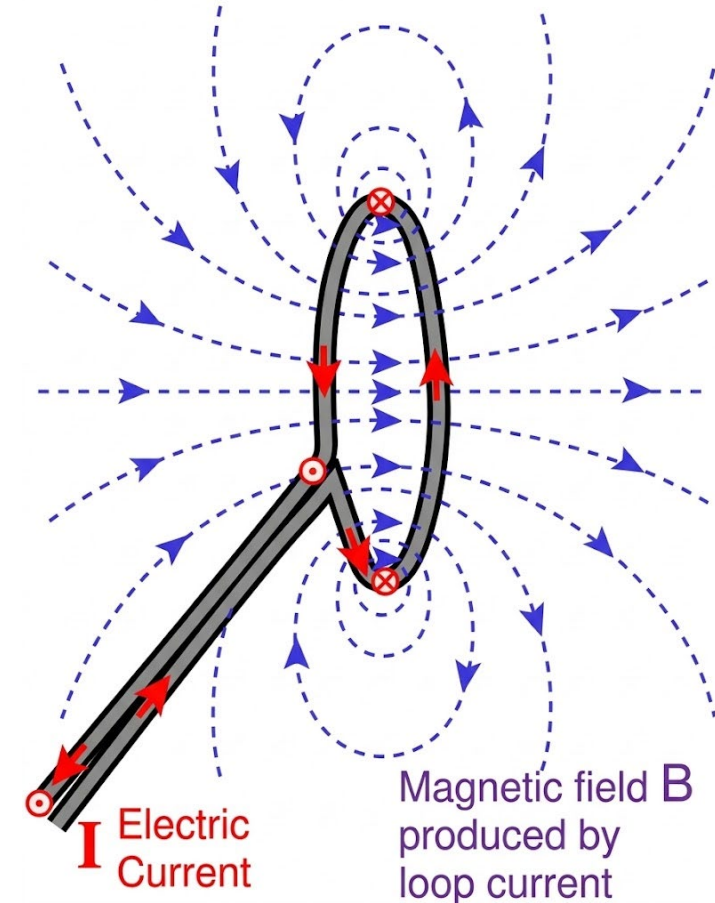
Total flux passing through the loop,

$$\Phi = \int_s \vec{B} \bullet d\vec{s} \quad [\text{Wb}]$$

This flux is also proportional to the current I .

$$\Phi \propto I \rightarrow \Phi = LI \rightarrow L = \frac{\Phi}{I}$$

The proportionality constant L is the self-inductance of the loop.

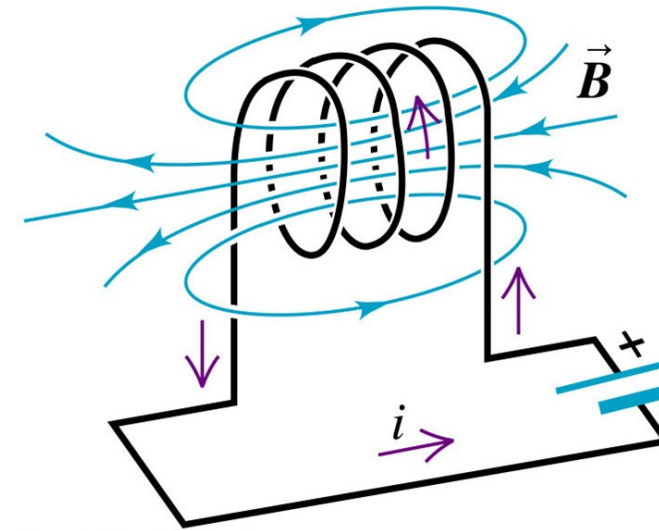


● Inductance measures magnetic energy storage capability.

Inductance in a N turn loop

Total flux that links the coil is proportional to the current.

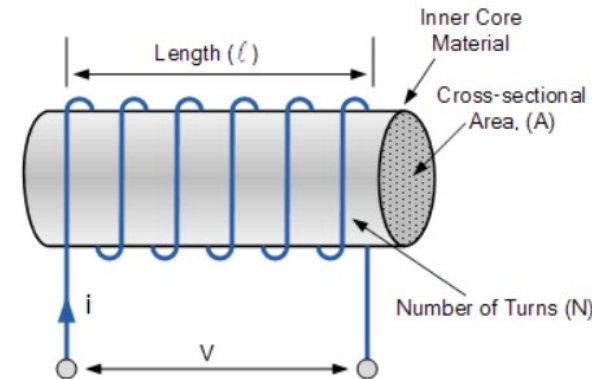
$$L = \frac{N\Phi}{I} = \frac{\lambda}{I} \text{ [Henry]}$$



The term $\lambda=N\phi$ is called *flux linkage*. (i.e., the flux seen by the coil turns.)

If there is the *magnetic saturation*, inductance becomes a function of current and the value may not remain constant. The most precise definition of inductance :

$$L = \frac{d\lambda}{di} = \frac{Nd\Phi}{di}$$



How to Calculate Inductance (Exam Method)

$$B \rightarrow \lambda (= N\phi) \rightarrow L (= \frac{\lambda}{I})$$

❶ Find B using Ampere's law $\oint_c \vec{B} \cdot d\vec{l} = \mu I_{\text{enclosed}}$ ● Always check: Is B uniform?

❷ Integrate to find Φ

$$\lambda = N \int_s \vec{B} \cdot d\vec{s} \rightarrow N\phi.$$

❸ Multiply by N $\rightarrow \lambda$

❹ Compute $L = \lambda/I$

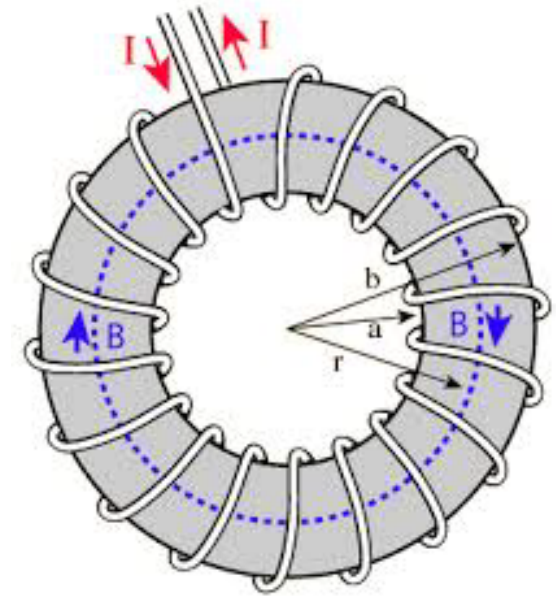
$$L = \frac{N\phi}{I} = \frac{\lambda}{I}$$

Inductance calculation from magnetic circuit

Recall flux of an unsaturated magnetic circuit can be found as,

$$\phi = \frac{NI}{\mathfrak{R}} \quad \text{where, reluctance } \mathfrak{R} = \frac{l}{\mu S}$$

$$L = \frac{\lambda}{I} = \frac{N\phi}{I} = \frac{N^2}{\mathfrak{R}}$$

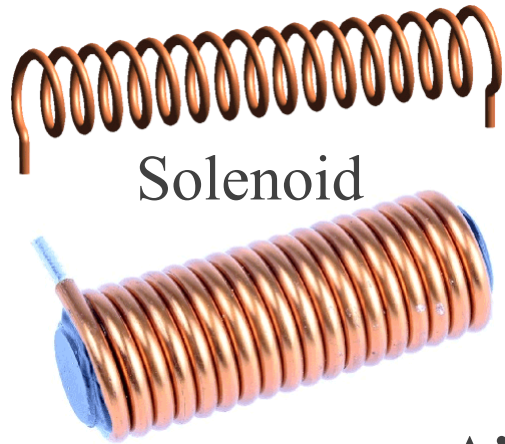


If the reluctance of a magnetic circuit is known, inductance L can be calculated directly (assuming no saturation and uniform flux density).

You have found reluctance of a toroid in example 2, now you can also find inductance of this toroid.

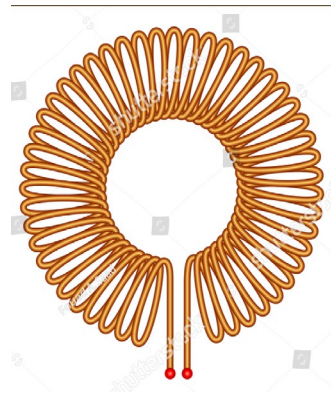
Inductors

When a coil, a solenoid and a toroid are designed specifically for their inductance to use as a circuit element, they are called **inductor**. (*Similar to this, the parallel-plate capacitors are specifically made for their capacitance and used as a circuit element.*)

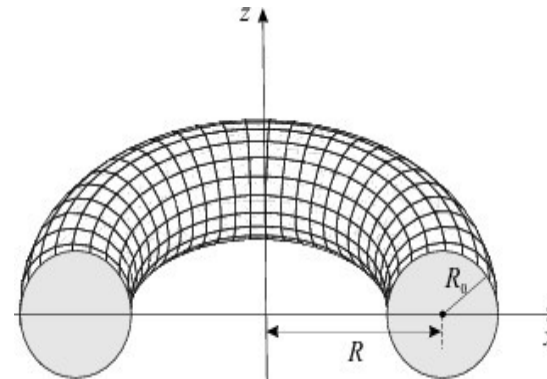


Solenoid

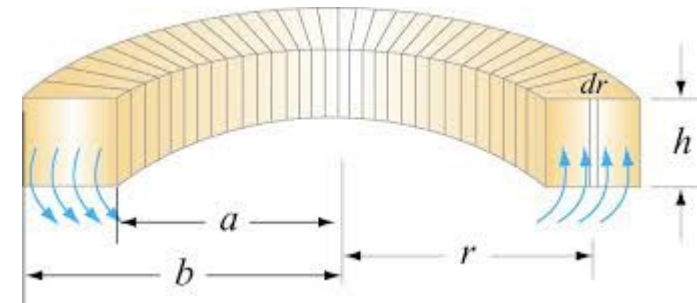
Air and magnetic core



Toroid



Round cross-sectional area



Rectangular cross-sectional area

How to calculate/design such inductors?

Example 5 (solenoid inductor)

Find the self-inductance per unit length of a long solenoid with $n=100$ turns per unit length and a diameter $d=50\text{mm}$. Assume air core.

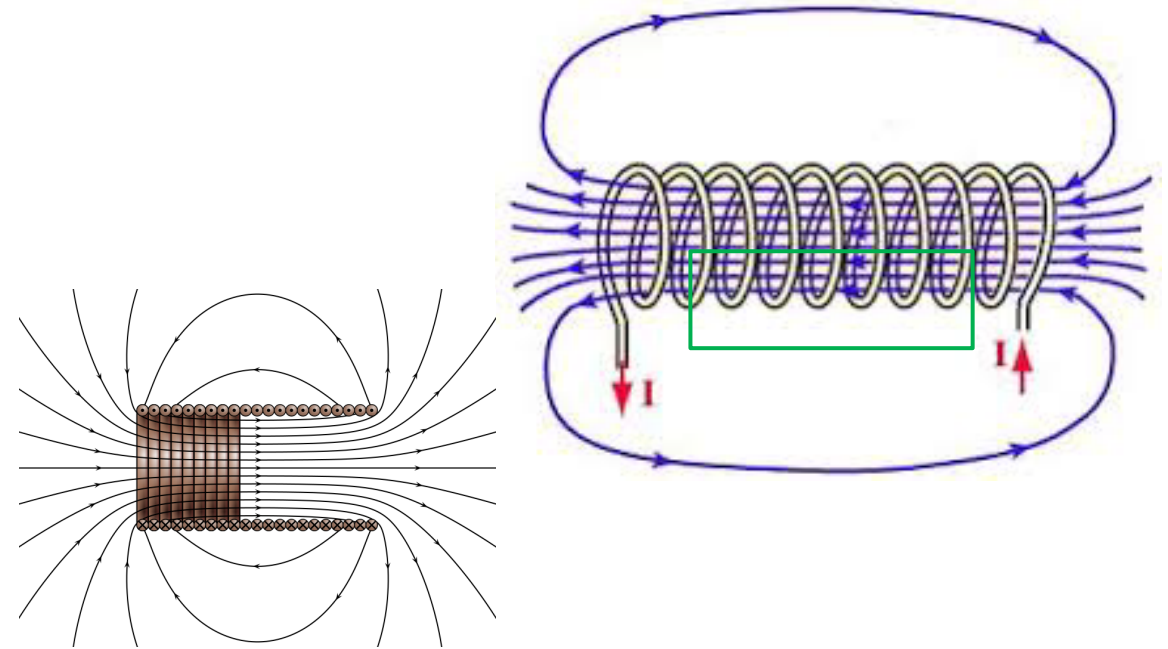
It is more useful to find inductance per unit length [H/m] for long solenoids (also for long cables and transmission line).

Solution:

Step 1: Find flux density from Ampere's law:

$$\oint_c \vec{B} \cdot d\vec{l} = \mu I_{\text{enclosed}} \rightarrow Bl = \mu_0 (nl) I$$

$$B = \mu_0 n I$$



Step 2: Find flux and flux linkage:

$$\phi = \int_s \vec{B} \cdot d\vec{s} \rightarrow \phi = BS = \mu_0 n I \left(\frac{\pi d^2}{4} \right)$$

$$\lambda = n\phi = \mu_0 n^2 I \left(\frac{\pi d^2}{4} \right)$$

$$\lambda = \mu_0 n^2 I \left(\frac{\pi d^2}{4} \right)$$

Step 3: Calculate inductance (per unit length),

$$L = \frac{\lambda}{I} = \mu_0 n^2 \left(\frac{\pi d^2}{4} \right) \quad [\text{H/m}]$$

● For long solenoid: $L = \mu n^2 S l \text{ [H]}$

- $L \propto \mu$
- $L \propto n^2$
- $L \propto \text{area}$
- Independent of current (if linear)

Can you find the same expression of L per meter using reluctance?

$n=100$ turns per meter and $d = 50$ mm

$$\therefore L = 24.7 [\mu\text{H/m}]$$

Can you design a solenoid (i.e., find d) of 3 mH/m with 10 turns per meter using (i) air-core, (ii) a magnetic core of relative permittivity 5000. Compare them and list the pros and cons for each?

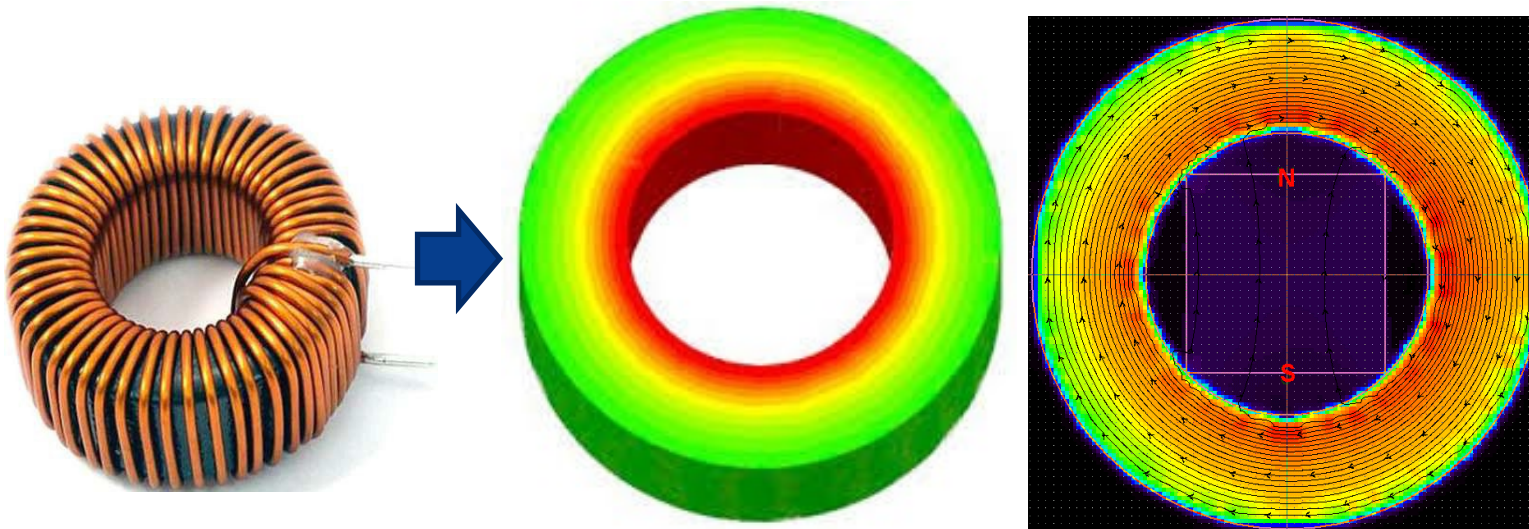
Example 6 (toroidal inductor)

A toroid of rectangular cross-sectional area has an inner radius a , outer radius b and height h . The coil has N number of turns and carrying current I . Calculate the inductance. Consider a non-uniform flux density in the core.

With an arbitrary radius r in the toroid core,

$$\oint_c \vec{B} \cdot d\vec{l} = \mu I_{enc} \rightarrow 2\pi r B = \mu(NI) \therefore B = \frac{\mu(NI)}{2\pi r}$$

B is not uniform everywhere. B varies as $1/r$ inside the toroid.



Flux density and lines in the core of the toroid

- Uniform B assumption $\rightarrow$ approximate method: calculation with reluctance
- Exact integration $\rightarrow$ logarithmic expression

Since B is not uniform everywhere, to find total flux consider an elementary cross sectional area ds at arbitrary radius r and integrate it over inner to outer radius to find the total flux.

$$\Phi = \int_s \vec{B} \cdot d\vec{s} = \int_s \left(\frac{\mu(NI)}{2\pi r} \right) \cdot (h dr) = \frac{\mu(NI)h}{2\pi} \int_a^b \frac{dr}{r} = \frac{\mu(NI)h}{2\pi} \ln \frac{b}{a}$$

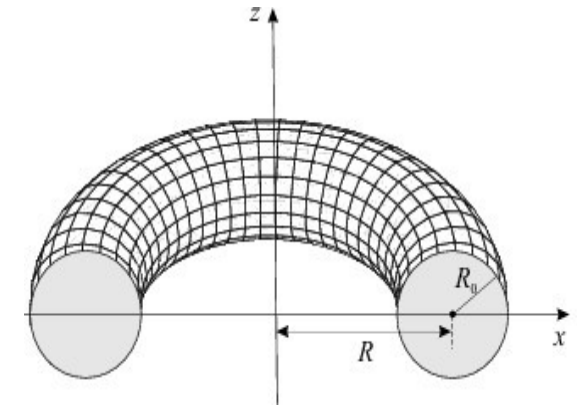
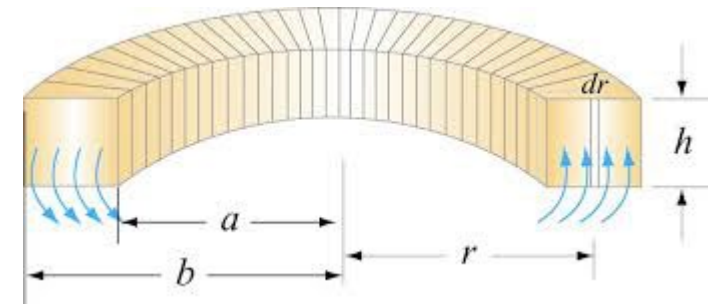
Flux linkage: $\lambda = N\Phi = \frac{\mu(N^2 I)h}{2\pi} \ln \frac{b}{a}$

Inductance: $L = \frac{\lambda}{I} = \frac{\mu N^2 h}{2\pi} \ln \frac{b}{a}$

● Standard result for rectangular toroid.

Compare this expression with that can be found from $L = \frac{N^2}{\text{Reluctance}}$

Can you find the inductance of a toroid with a circular cross-sectional area but non-uniform B ?



Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.



0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud

All responses

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1 of 3

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Scan the QR or use link to join



<https://forms.office.com/r/vJs4eS-WdQH>

 Copy link

Key Summary points

① Magnetic Circuit Model (Linear Region Only)

$$\Phi = \frac{NI}{\mathfrak{R}}, \mathfrak{R} = \frac{l}{\mu S}$$

- MMF (NI) drives flux
- Reluctance opposes flux
- Analogy to Ohm's law
- Valid only if μ is constant (no saturation)

② Air Gaps Matter Most

$$\text{Since: } \mathfrak{R}_g = \frac{l_g}{\mu_0 S}$$

$$\text{and } \mu_r \gg 1, \mathfrak{R}_g \gg \mathfrak{R}_c$$

- Air gap usually dominates total reluctance.

③ When Core Saturates

- μ is no longer constant
- Magnetic circuit method fails
- Use **load-line method**

Inductance Definition: $L = \frac{\lambda}{I}$

where: $\lambda = N\Phi$

- 1 Use Ampere's law $\rightarrow$ find B
- 2 Integrate $\rightarrow$ find flux Φ
- 3 Multiply by N $\rightarrow$ flux linkage
- 4 Compute $L = \lambda/I$

Key Results to Recognise

For long solenoid: $L = \mu n^2 A \ell$

For rectangular toroid:

$$L = \frac{\mu N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$$

Design Insights

- $L \propto \mu$
- $L \propto N^2$
- $L \propto$ cross-sectional area
- Air gap reduces L dramatically

● Exam Signposting

What You Must Be Able to Do

■ Core Computational Skills

- ✓ Apply Ampere's law correctly
- ✓ Identify correct symmetry
- ✓ Calculate reluctance
- ✓ Compute flux from MMF
- ✓ Calculate inductance

■ Conceptual Understanding

- ✓ Explain difference between B , H , and M
- ✓ Explain why air gap dominates
- ✓ Explain why saturation invalidates magnetic circuit method
- ✓ Interpret a B – H curve

■ Typical Exam Question Types

Compute flux in magnetic circuit with air gap

Compare two designs (air-core vs magnetic core)

Determine operating point using load-line

Derive inductance of solenoid or toroid

● Common Mistakes to Avoid

- ✗ Forgetting air-gap reluctance
- ✗ Assuming B uniform when it is not
- ✗ Using constant μ in saturation region
- ✗ Forgetting N^2 in inductance

Review and further practice (Recommended)

Practice Example 6-10, 6-11, 6-12, 6-13 from the textbook by D.K Cheng